Chapter 2 Challenge: Answer Key

Rational Numbers: Challenge Set (show all work)

1. -0.708333... = -0.708 with 3 repeating

2. -4 1/16

3. Let x = 0.3636..., so 100x - x = 36 and x = 36/99 = 4/11

4. -138/36 + 100/36 + 45/36 = 7/36

5. (-7/3)(9/5) = -21/5, then (-21/5)(-10/3) = 14

6. -0.3 + (-0.94) = -1.24

7. 9/16 - (5/8 / -1/8) = 9/16 + 5 = 5 9/16

8. Rises 25.5 ft to -17 ft, then -17 - 18.75 = -35.75 feet

9. Sum = -1.325, so the average is -1.325 / 4 = -$0.33125 (about -$0.33) per day

10. 14 / (8/3) = 5 1/4, so 5 full batches, with 14 - 40/3 = 2/3 cup left